

Forensic Science

for High Schools

Book 2

John Powrie and Jeannette Jolley



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Forensic Science

Book 2

John Powrie and Jeannette Jolley





Emerald City Books

First published 2000

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Printed in Australia by Star Printery

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Emerald City Books
An imprint of Blake Education Pty Ltd
A.C.N. 074 266 023
Locked Bag 2022, Glebe NSW 2037, Australia

ISBN 978 1 92571 070 0



Forensic Science Book 2 takes a detailed look at two topics in the field of forensic science. Topic 1 examines the subject of fingerprinting; some historical background is combined with theory and activities where the students can take their own fingerprints, while Topic 2 looks at the detection of forgery on paper and metal, using techniques such as electrostatic imaging and handwriting comparison. Forensic Science Book 1 provides an introduction to forensic science and a study of contact traces.

Each book in the series contains factual information on several distinct subject areas which is interspersed with interesting practical activities and revision questions. The books provide units of work with the flexibility to allow teachers to personalise subjects to suit their classes. The activity worksheets require science equipment, most of which should be available in the school laboratory, such as Bunsen burners, petri dishes, beakers, hand lenses, glass slides and filter paper. The teacher should read through activity sheets beforehand for specific materials needed. The majority of the worksheets are self-contained.



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1 FINGERPRINTS

Fingerprints are the marks left behind when someone touches an object with their fingers. There are three kinds of fingerprints:

- an impression left in something soft, such as putty, soap or wet paint
- a print left by a hand covered in dirt, blood or paint
- an invisible deposit left by the secretions from the skin (small amounts of oil and perspiration are secreted from microscopic pores on the tiny ridges on the fingertips and other parts of the body).

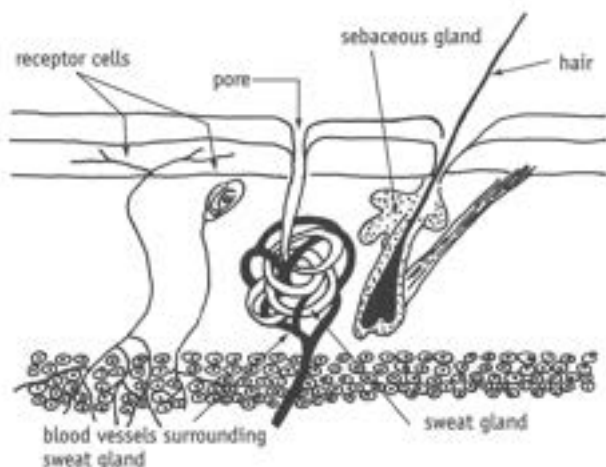


Fig. 1 Sweat glands secrete water, amino acids, salts and urea via the pores. Oil produced by the sebaceous glands also oozes onto the skin surface. While the water evaporates, the salts, oil, urea, amino acids and skin debris are left behind in a fingerprint.



Fig. 2 This diagram shows the minute pores in the ridges of the skin through which the sweat droplets emerge.

When an object is touched, a small amount of these secretions is transferred onto that object in exactly the same pattern as the ridges on the fingers or palm that made the contact. It is this pattern that makes the distinctive fingerprint pattern.

Every person has their own unique pattern of ridges and furrows. They are found on the outer layer of skin of fingers, thumbs, toes, lips, palms and soles of the feet. There is even a record of a lip-print used to identify a car in a 'hit and run' accident in the United States. No two people have the same pattern of ridges, not even identical twins. These patterns are completely formed by the fifth month of pregnancy and will never change in shape throughout the person's lifetime. After most injuries and burns the same patterns will return as the skin heals. Fingerprints are only altered by the most severe of injuries. Then, the injury itself may become as unique and distinctive as the fingerprint it has replaced.



Questions

1. Make a list of the three types of fingerprints that may be found at a crime scene and, for each type, write a sentence describing the type of crime and setting where such a fingerprint may be found.
2. Which parts of the body display ridge patterns that allow an impression to be made of them?
3. What are the two glands found in the skin, and what does each one produce?
4. Describe what an invisible fingerprint could be made of, and explain how such a fingerprint is produced on a hard surface.
5. Make a simple copy of the diagram of the skin above, labelling all the main parts.
6. When are a person's fingerprints formed and how may they change or be changed?
7. How many people are likely to have the same fingerprint pattern?



There are three main patterns of fingerprints.



Arch



Loop



Whorl

Fig. 3

To help identify a fingerprint as one of these patterns, a triangular shape formed where two patterns meet was recognised and named a **delta**. The original three basic patterns were later expanded into the following eight groups using the idea of the delta to help in classification. The **core** of the fingerprint, found in the centre of the pattern, is a 2nd fixed point used to help in classification.

An **arch** was a pattern that ran from one side of the finger to the other without making a backward turn. Hence an arch has no delta. A **tented arch** has an upward ridge that gives the pattern a spine.

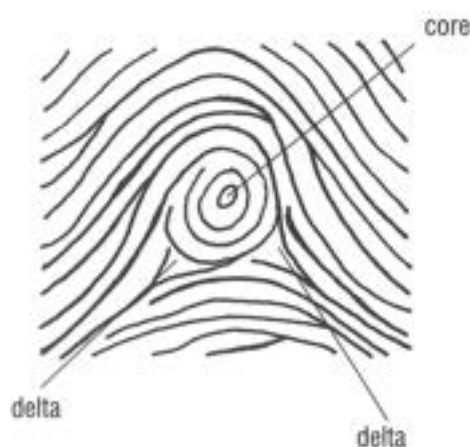


Fig. 4



Tented arch

Fig. 5

Identifying fingerprints

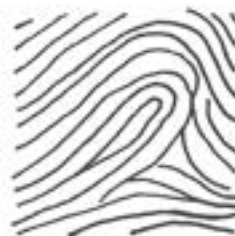
1 FINGERPRINTS



When the ridge makes a backwards turn a **loop** is formed. A loop has 1 delta. A loop where the ridges slope downwards towards the thumb is called a **radial loop**, and a loop where the ridges slope towards the little finger of the hand is called an **ulnar loop**.



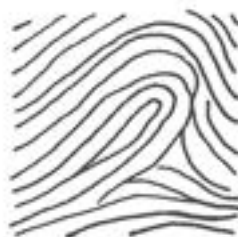
Radial loop on left hand



Ulnar loop on left hand

Fig. 6

A **whorl** is a pattern with 2 *deltas* where the ridges form a complete turn. A central pocket loop has two deltas. It has a *core* like a whorl, but the rest of the pattern is like a loop. A double loop has two clear intertwined loops, with two *deltas* and two *cores*.



Central pocket loop



Double loop

Fig. 7

Composite fingerprint patterns are not common. They consist of a mixture of patterns, and may have three or more deltas. These patterns are a complex system of whorls and loops.



Composite patterns

?

Questions

1. Make a simple sketch of the eight main fingerprint patterns, naming each type.
2. Explain what the core and delta are, and label them on the diagrams you have drawn.
3. Make a list of the three types of patterns that have one delta.
4. Describe how the patterns of a whorl and a pocket loop are different.
5. Explain how you would decide if a fingerprint were a composite. Sketch an example.

Fig. 8



Whenever a crime has been committed, fingerprints of all people known to have been legitimately at the scene of a crime need to be taken to find which prints may have been left by the criminal. Suspects are also fingerprinted so that they can either be investigated further, or be eliminated from further inquiries. These fingerprints are called **elimination fingerprints** and serve as a 'control' against which prints found at the crime scene can be compared.

The fingerprints are taken using a special ink pad. The fingers are each separately lightly rolled on the ink from one side of the nail to the other covering the whole of the first joint. They are then carefully rolled on paper on a special form in the place indicated for each finger. This is repeated for all 10 fingers. All fingers are then printed again together to check the order of the individual prints and the thumbs together for the same reason. Usually whole palm prints are also recorded. Rolled impressions are used because they give maximum detail that may be needed and show the full width of the print.

Your teacher may give you a copy of a typical form used by police departments to record elimination fingerprints; the Police Elimination Fingerprints sheet. These can then be compared to any left at the crime scene.

?

Questions

1. Which type of fingerprints do you think would be the clearest—elimination fingerprints, or those found at a crime scene? Explain your answer.
2. Under what circumstances would a person's fingerprints be taken by the police?
3. Why do you think palm prints are also usually recorded?
4. Miss Green had been collecting money for a class party, and had stored it in a cash box left in her desk. The desk was broken into, and the money taken. Some students mentioned that they saw some students from another class acting suspiciously around the classroom at recess, and were able to give the police a good description of those involved. Four suspects were soon found. Their fingerprints and those of Miss Green were taken. The police were also able to get three good thumbprints (A, B and C) from the cash box. The suspects' thumbprints are shown below.

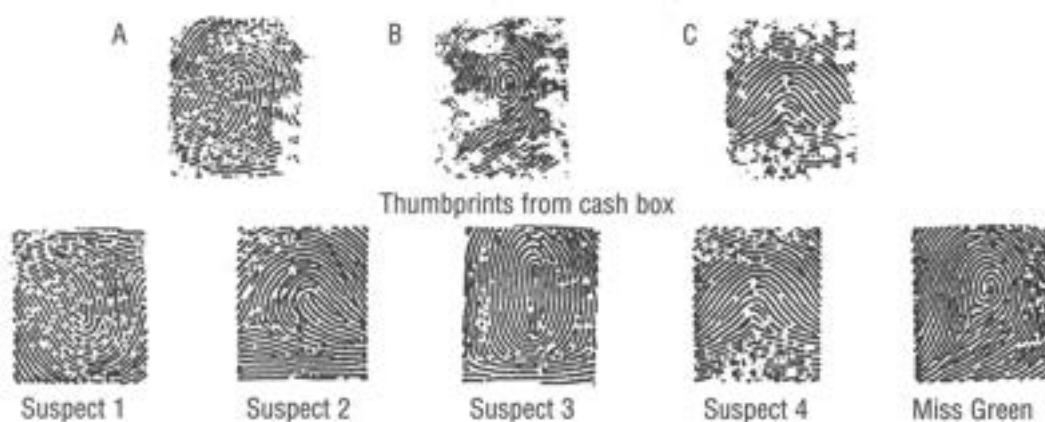


Fig. 9

Suspect's thumb prints

- (a) Which of the thumbprints on the cash box belonged to Miss Green?
- (b) Did any of the suspect's thumbprints match thumbprints found on the cash box? Does this mean that this suspect stole the money? Explain your answer.
- (c) Explain to whom the third thumbprint on the box may belong.



In this activity you will make a record of your own fingerprints in much the same way the police take those of criminals, suspects and witnesses. It is a good idea to practice a few times on scrap paper first until you are confident you can get a print which shows the pattern of ridges clearly.



Materials

- newspaper and scrap paper
- ink pad
- soap and towel
- form or paper on which to make the final print
- hand lens.

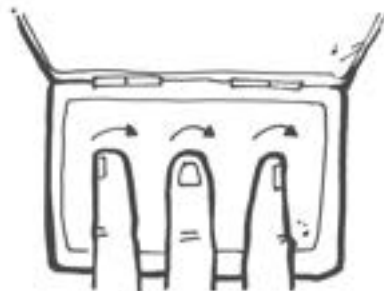


Procedure

1. Place the newspaper on the bench with the ink pad and your record sheet on top of it.



2. Placing your hand near the edge of the bench, roll the end joint of your first finger on the ink pad so that the whole joint is inked.



3. Being careful not to use too much ink, which will give a blurred print, roll your finger once only from side to side on to your record sheet.



4. Wipe your finger, then repeat with all your fingers and thumb from both hands, making a print of each on your record sheet.



5. Once you have all your fingerprints recorded, wash your hands well with soap, read the information that follows and answer the questions.

Race and fingerprint pattern

Earlier this century it was found that in Western European populations the loop was the most common type of fingerprint, accounting for about 60% of all prints. Arches were quite rare, being found in only about 5% of all prints, with whorls and composites making up the last 35% of fingerprints. Loops are also the most common pattern among black Africans and the Ainu people of Japan.

Activity: Taking your own fingerprints

1 FINGERPRINTS



It has been found that among the African Bushmen and Central Europeans, arches are the most common fingerprint, while whorls are more common in the Australian Aboriginal and Mongolian populations. There also appears to be a weak link between a tendency to have high blood pressure and more whorls as well as a link between different blood groups and fingerprint patterns.

?

Questions

1. Copy the following table into your science book. Using a hand lens, try to identify the patterns on all ten fingers. Record the total number of each type of pattern in the table.

Pattern	Arch	Loop	Whorl	Composite
Myself				

2. Compare the patterns from your fingers with those of a friend. Are there any similarities?
3. Carry out a class survey and see which patterns are the most common. Add an extra line to the bottom of your table and record the numbers of each pattern in the class.
4. How do these results from your class compare to the figures for the general population? Can you suggest why they may not conform exactly with the national figures?
5. Using your results from the class survey, draw a histogram showing the proportion of each pattern in the class.
6. Using your graph, write a summary of your results, describing how common or rare each pattern is in your class.
7. Read the paragraph below on how fingerprints form. In your science book write a paragraph explaining how the papillary ridges form, and how the foetal pads determine the type of pattern that occurs.

How the fingerprint patterns form

A person's fingerprints form while still in the womb in the third or fourth month after conception. Before this stage, the foetus has small swellings on the ends of all its fingers and toes. These are called foetal pads. As they become smaller, the outer layer of skin shrinks and wrinkles slightly to form the papillary ridges and furrows that will become the fingerprint pattern.

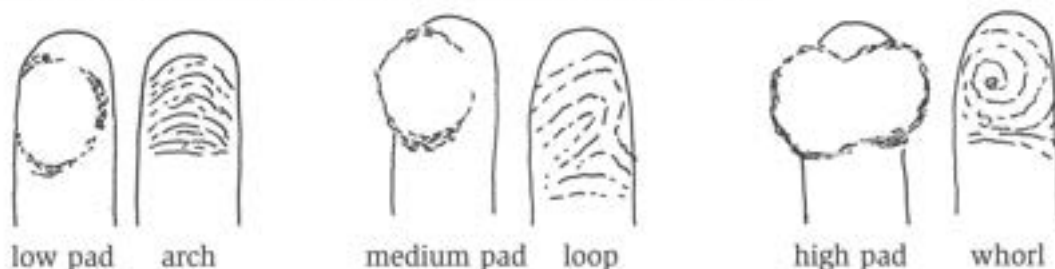


Fig. 10

It has been found that the larger swellings usually form whorl patterns, the medium swellings, loops, and the smallest swellings form arches. The height of the swelling also determines the number of ridges as well as the pattern type.

Once formed, fingerprints only grow in size as the body grows: the patterns never change, except in the case of severe injury to the lower layers of the skin.



POLICE ELIMINATION FINGERPRINTS

THESE FINGERPRINTS WILL BE DESTROYED AT THE CONCLUSION OF THE CASE

Name _____ Prints taken by _____

Address _____ Rank _____

Division _____

Offence _____ Date _____

ROLLED IMPRESSIONS OF RIGHT HAND

1. R. THUMB	2. R. FOREFINGER	3. R. MIDDLEFINGER	4. R. RINGFINGER	5. R. LITTLEFINGER

ROLLED IMPRESSIONS OF LEFT HAND

6. L. THUMB	7. L. FOREFINGER	8. L. MIDDLEFINGER	9. L. RINGFINGER	10. L. LITTLEFINGER

PLAIN IMPRESSIONS OF BOTH HANDS

Impressions of the four fingers LEFT HAND	Thumb impression LEFT HAND	Thumb impression RIGHT HAND	Impression of the four fingers RIGHT HAND

1-5

CONTINUED

Activity: Taking your own fingerprints

1 FINGERPRINTS



LEFT PALM

RIGHT PALM



The earliest records of fingerprinting are by the **ancient Chinese**, who used thumbprints to seal important documents. In 1684, **Dr Nehemiah Grew**, a British scientist who specialised in studying the structure of plants, published a paper that included drawings of various fingerprint patterns. About the same time, **Marcello Malpighi**, an Italian doctor and professor of anatomy, was involved in research on the function of skin. He also described the different patterns made by the papillary ridges.

The first record of fingerprints being used as a means of identification did not occur until 1858. **Sir William James Herschel** was a British administrator in the province of Bengal in India. He was frustrated by not being able to distinguish between men who were impersonating others to collect multiple pensions. Herschel solved the problem by asking illiterate men to place their thumbprint on the pension receipt, in the place of a signature.

Not long after Herschel's work in India, a Scottish doctor, **Henry Faulds**, who was working in Tokyo, Japan, was also investigating fingerprints. He published an article in the scientific journal *Nature* in 1880 in which he discussed the pattern of ridges and suggested its use in the identification of criminals. The work by these men attracted the attention of an English scientist, **Francis Galton**, who in 1890 was able to acquire Herschel's research material. After his own investigations, Galton concluded that fingerprints were indeed unique to each individual. He went on to calculate that the chance of two different fingers having exactly the same print was 1 to 64 000 million, which is greater than the whole earth's population. He was the first person to recognise the importance of the small triangle region later called the *delta*.

So, the acceptance of using fingerprints to identify one individual was established, but the stumbling block was the lack of a simple classification system. What was needed was a system under which all fingerprints that were taken could be filed and then retrieved to be compared against any found at a crime scene. In Argentina, a police officer named **Juan Vucetich** was working along similar lines, and in 1892 he was able to secure the world's first conviction using fingerprints as evidence.

An Englishman, **Edward Henry**, is credited with having created a system on which the present method of classification and filing is still based. Henry was Inspector General of Police in Bengal in India where he saw both the advantages and drawbacks of using fingerprinting. While on leave in England he contacted Galton who passed on all his research data. Back in India, Henry set to work. He recognised five basic patterns—the three (*arch*, *loop* and *whorl*) identified by Galton, but added the *tented arch* and the *double loop*. He introduced the idea of fixed points, the *core* and *delta*, and used these to group these patterns into four main groups according to the number of deltas (none, one, two or more than two). By 1910 the '*Henry System*' of classification of fingerprints was used throughout most of the world.

Fingerprints were stored in large filing cabinets and searched for manually. When a possible match was found they were then compared using a hand lens. This took a lot of time and detectives had to be patient and have a good eye for detail. Looking for a matching fingerprint could take days. Nowadays, fingerprints are stored on computer databases, speeding up the process. However, the system of classification first developed by Edward Henry is still the basis of the system of classification that is used in all English-speaking and most European countries today.



Questions

1. In your books, draw up a timeline showing all the major advances in the use of fingerprints in forensic science, who was responsible, where this occurred and what they contributed.
2. Do some research in your library to add the following names to this timeline: Jan Evangelista Purkinje (1787 – 1869) and Alphonse Bertillon (1853 – 1914). Include details of their work.
3. Of the scientists discussed above, which do you think made the greatest contribution to the development of the fingerprinting technique? Explain why you believe this.



Even considering the eight main fingerprint patterns, it would be impossible to declare two fingerprints to be from the same finger if you were relying purely on matching patterns. So, when matching two fingerprints, it is not the patterns that are important but the characteristics, the little variations that occur in the ridge patterns. For two fingerprints to have been made by the same finger, they must have exactly the same characteristics in exactly the same place.

The diagrams below show eight examples of such characteristics.

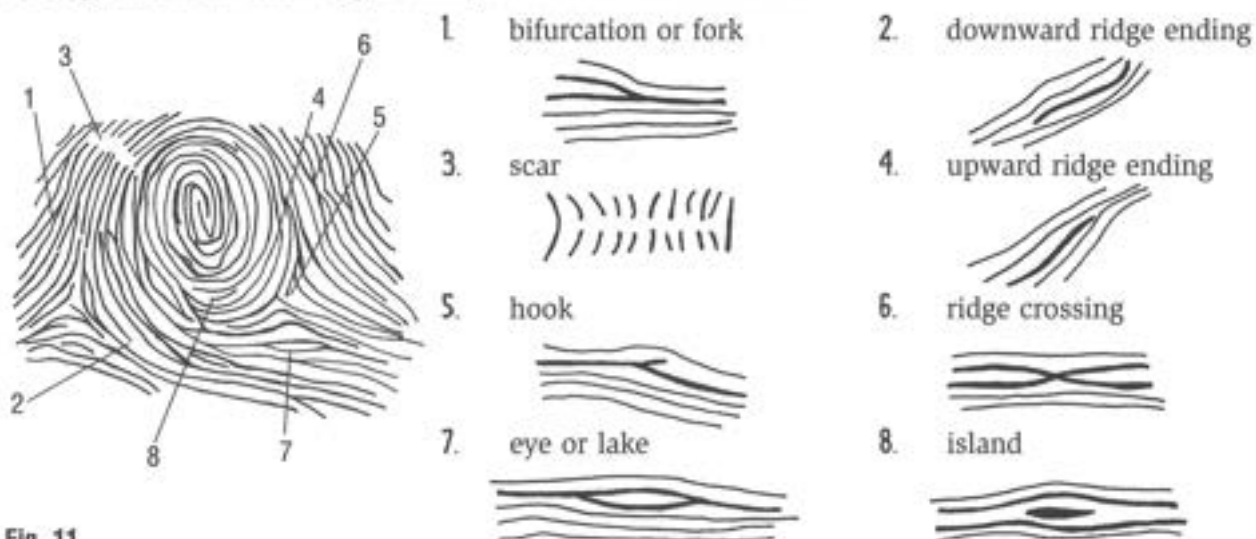


Fig. 11

In some countries there must be a certain number of characteristic points in common if two fingerprints are to be accepted as a match. However, in most parts of Australia there is no minimum number of points of identity required. Instead, a conviction depends on all the available evidence. If a recognised fingerprint specialist declares that, in their opinion, two prints are a match, then legally they are accepted as a match. Of course, the greater the number of matching characteristics, the stronger is the print as evidence. Clearly, there needs to be only one single, unexplained difference between two prints for them to be declared not a match.

?

Questions

1. Explain what a characteristic point is, and how a fingerprint expert would use one to try to match a fingerprint found at a crime scene with one on file.
2. Below are two fingerprints that look the same. Try to find at least ten possible characteristics in common and label them.
3. Would you say that the two fingerprints are an exact match? If so, explain why you have come to this conclusion.
4. Explain why one fingerprint is a square shape and the other more rounded.



Fig. 12



Fingerprints made by hands covered in blood, soot or dirt are clearly visible and can be easily photographed and compared for a match. However, most fingerprints are not obvious. The pattern of salts and oils left behind when the water in the perspiration has evaporated is not easily seen. These are called latent fingerprints and they have to be developed in some way to make them visible so that they can be photographed. The method used depends on the type of surface on which they are found.

1. Dusting for fingerprints

If fingerprints are on a hard or non-porous surface such as glass, glossy paper or metal, they can be simply dusted with a suitable powder and photographed. The powder sticks to the moist parts of the impression and the loose excess powder is blown or gently brushed away. If the background surface is dark, a white or light powder such as powdered aluminium or talc is used. On white or light surfaces a dark powder such as carbon is used instead.



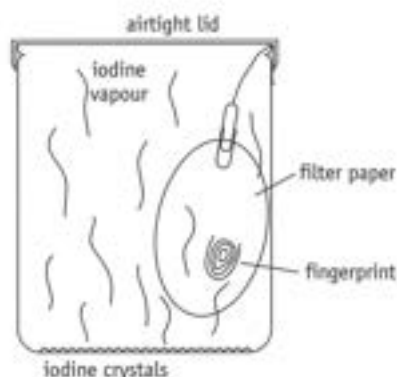
After photographing, the dusted print can be lifted with some clear sticky tape. The tape is placed over the print and the powder lifted from the surface. It is then fixed on to paper of a contrasting colour. This is just an extra precaution in case the photograph is not clear enough. It can then be kept as a permanent record, and compared to other fingerprints already on file.

2. Fingerprints on absorbent surfaces using iodine (teacher demonstration)

Often a fingerprint leaves oil and perspiration on an absorbent or porous surface, such as cloth or non-glossy paper, that soaks up the fluids, making them unsuitable for dusting. Such latent prints can still be made visible by other means such as iodine vapour. This should be done as a demonstration by the teacher, because iodine is corrosive, toxic and a harmful irritant. The method works because iodine fumes are absorbed by the oil in the impression and leave a brown pattern on the surface.

Place a firm fingerprint on a small piece of filter paper. Attach a cotton thread to one end of a paper clip and then attach the paper clip to the filter paper, being careful to not add any further prints to it.

Lower the filter paper into a screw-topped jar containing some iodine crystals, as shown in the diagram on the right. Close the lid tightly. If it is a cold day, speed up the sublimation of the iodine crystals into iodine vapour by lowering the jar into warm water. Watch carefully so that you can record your observations.





Because iodine sublimates readily in air, the developed fingerprint must be sealed. This can be done by enclosing it completely in sticky tape, or by lowering it into a solution of calcium chloride and potassium bromide for a few minutes.

Today, police often use an aerosol spray of Ninhydrin to develop latent fingerprints on porous surfaces. This chemical reacts with the amino acids left behind in fingerprints. Impressions sprayed with Ninhydrin show up as pink to purple but change to a deep blue when exposed to ammonia.

X

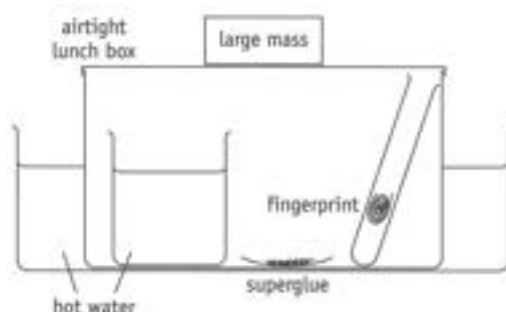
Exercise

Write a short report in your science book on what you observed. Record how long it took for the fingerprint to become visible, what it looked like, and what happened to any developed print that was not sealed. Describe how the print was sealed, and how successful this was.

3. Making fingerprints visible using superglue (teacher demonstration)

Superglue is frequently used at crime scenes to show up fresh latent prints on smooth non-porous surfaces. In the presence of water vapour, the superglue leaves a clear white deposit in the pattern of the fingerprint. This activity is best done as a teacher demonstration, as superglue can be quite dangerous if it gets on to the skin.

Place a test tube or microscope slide with a fresh fingerprint on it in a plastic lunchbox, as shown in the diagram on the right. In the same box place a small beaker half-full of hot water and a watch glass containing ten to 15 drops of superglue. Then, seal the lunchbox and place it in a large water bath of hot water to ensure a humid atmosphere inside the lunchbox. You can place weights on top of the lunchbox to keep it down in the water bath. After about half an hour open the lid, preferably in a fume cupboard, and you should be able to see a clear white pattern. Fingerprints developed this way cannot be easily lifted but are usually clear enough to be photographed.



X

Exercise

Write a report on what you observed and include a scientific drawing of the apparatus used.

4. Fingerprints on sticky surfaces (teacher demonstration)

If fingerprints have been left on a sticky surface such as sticky tape, a bottle or glass, dusting will not be an appropriate method of revealing such a print, as any dust used would also adhere to the sticky surface as well. In these cases, another chemical that will remove the sticky surface is used. The chemical is a solution of gentian violet and hydroxybenzene (phenol) crystals in ethanol. This solution is toxic, flammable and corrosive so care must be taken.



Procedure

1. Using about eight cm of sticky tape, stick one end around a glass rod and make a fingerprint on the other end.



2. Place the whole of the tape in the gentian violet solution for about 15 minutes. Then, rinse under tap water to remove the sticky surface. Observe the visible fingerprint.



Questions

1. What is a latent fingerprint? Explain what has to be done to make them useful in a police investigation.
2. Why are dusted fingerprints lifted if they have already been photographed?
3. With what part of the fingerprint does iodine react?
4. Explain why an iodine print needs to be sealed, and describe how this is done.
5. Describe the conditions required for superglue to develop a fingerprint. How is this done in the classroom?
6. What is the colour of the fingerprint left on the tape?
7. What are some sticky objects found at a crime scene that might have fingerprints left on them?
8. Why isn't it appropriate to dust such surfaces for prints?
9. Make a list of three places in which fingerprints may be found. Describe the method of developing and recording these prints in each case.
10. Give three examples of some places where fingerprints cannot be easily recorded.



In this activity you will make an invisible (latent) fingerprint visible by dusting it with a suitable powder.



Materials

- some glossy black paper or plastic
- a glossy white surface such as a tile
- some talcum powder
- some fine carbon powder
- a very fine brush such as a make-up brush
- newspaper.



Procedure

1. Place some newspaper on the bench to catch any excess powder.
2. Sprinkle some powder of a contrasting colour on the area that was fingerprinted.



3. Place the glossy black surface on the newspaper and make an impression with a finger or thumb.



4. Very gently brush off any excess powder to leave a visible image of your fingerprint. This can be lifted using sticky tape and placed in your science book.



Questions

1. On which surface were the fingerprints most clear?
2. Would this method of dusting prints be suitable for fabric or furry surfaces? Why, or why not?
3. Why would talcum powder be more suitable for dusting than, for example, sugar or salt?
4. What type of powder would you choose to dust fingerprints found on a black ceramic vase? Explain why you have made this choice.



Since the 1960s, all developed countries have stored the millions of fingerprints on their files on computer databases. The fingerprints are now converted into a digital image, and the computer calculates the positions, directions and relationships among the basic patterns. The computer compares the data on file with a fingerprint found at a crime scene and then produces the names of possible suspects in order of probability of the prints being a match. A fingerprint expert then needs to make a manual search of only a few prints.

Other advances in the matching of fingerprints include the use of lasers. This has been of particular help in situations where the prints at a crime scene have been weak, damaged or old.

The method involves turning the lights off, then 'painting' the room with a blue-green beam. When the light is directed on to a fingerprint it glows brightly, and can be photographed. Fresh fingerprints have a greenish-yellow glow, while older prints fluoresce orange.

No one is quite sure what chemicals in the prints react with the laser light to make it glow, nor why some prints will show up in laser light, but not by any other means.

The advantage of using lasers is that they are non-destructive. The method leaves no residue and won't damage the evidence. In fact, it is usually the first step in trying to find the best method of developing any prints. Today many forensic laboratories have small portable lasers with variable wavelengths that can be taken to the crime scene.

However, using lasers is very expensive: one laboratory-based laser costs more than \$50 000. An often used cheaper alternative is to dust the area with fluorescent dye, then illuminate it with UV light. If the scene is viewed through tinted glasses, the fingerprints will also glow.

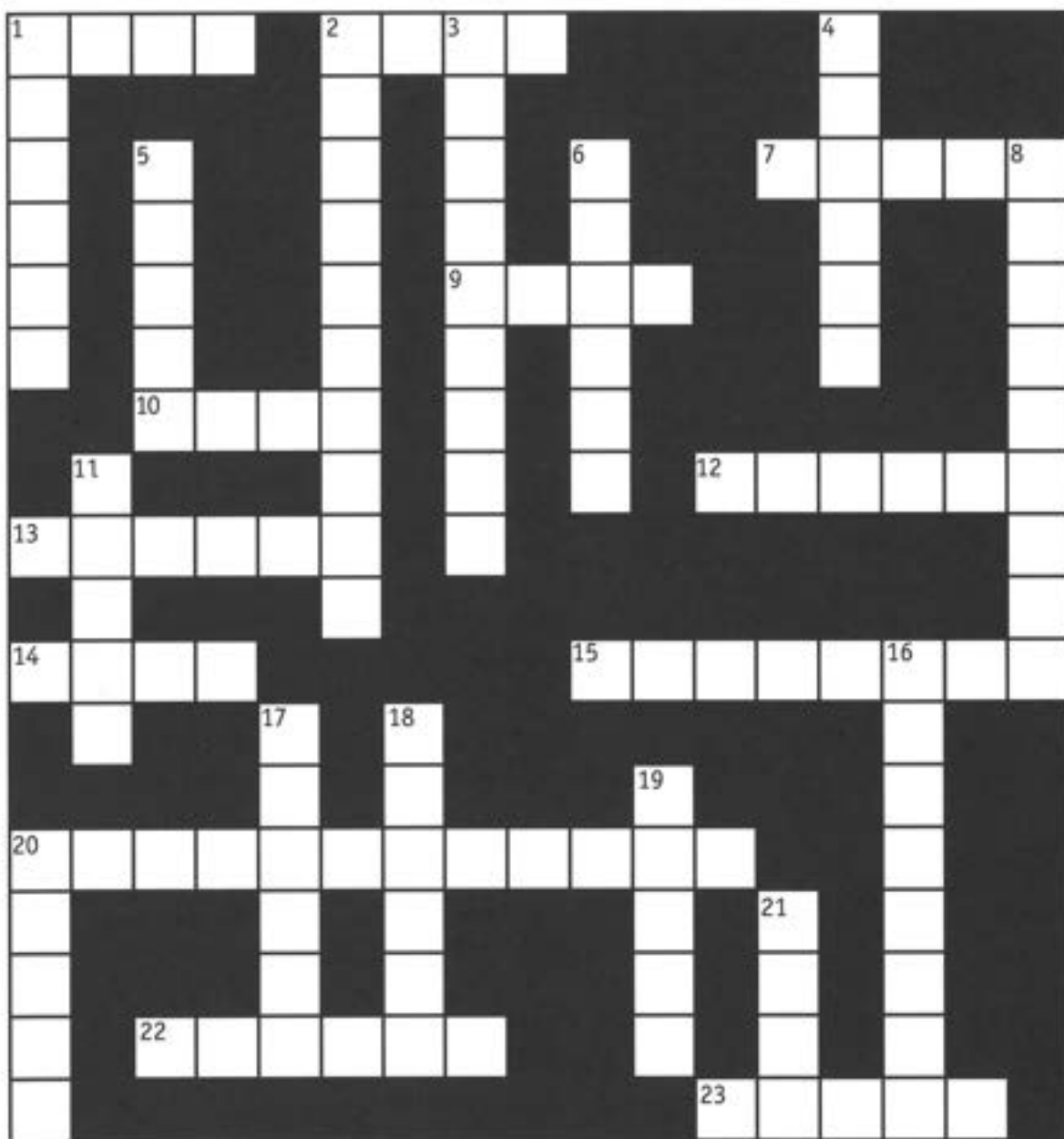
Since the 1970s, many new chemicals have been tested to help develop latent fingerprints. In fact there are now more than 40 accepted ways to develop latent fingerprints. Identifiable fingerprints have been developed on rough, absorbent paper and cloth, and even on the skin of a body.



?

Questions

1. What advantages are there in putting fingerprints on a computer database?
2. Explain how lasers have improved the detection of latent fingerprints.
3. Describe how fingerprint detection has improved since the 1970s.
4. Why is fingerprint detection so important in forensics?
5. What further developments could make fingerprint detection even more effective?





Across

1. Number of basic patterns first recognised by Edward Henry.
2. Fingerprint pattern that has no deltas.
7. Most common fingerprint pattern.
9. Substance found on skin surface.
10. Fine white powder used for detecting latent fingerprints.
12. Used to detect fingerprints on absorbent surfaces.
13. Type of arch.
14. Dr Nehemiah studied fingerprints in the 17th century.
15. Iodine does this when it is warmed.
20. All fingerprints are in order to make a permanent record.
22. Name given to fingerprints that cannot be seen with the naked eye.
23. Country where William Herschel first used fingerprints to identify individuals.



Down

1. Scottish doctor working in Tokyo who first suggested the use of fingerprints to identify criminals.
2. Substance found in sweat.
3. Fingerprint containing more than one pattern.
4. Fingerprint pattern having two deltas.
5. Type of gland in skin.
6. Scientist who first identified the delta.
8. Gland in skin that produces oil.
11. Englishman who invented the basic method of classifying fingerprints that is still used today.
16. Italian doctor who studied skin and papillary ridges in the 16th century.
17. Gentian is used when looking for fingerprints on sticky surfaces.
18. A fine black powder used to detect fingerprints on light, smooth surfaces.
19. A triangular shape formed when the ridges of 2 different patterns come together.
20. Part of the body other than the fingertips that can be used for identification.
21. Argentine police officer, Vucetich, devised a method of classifying fingerprints.



Forgery is the changing of a document, or the making of an imitation of something, with the intent to cheat.

Types of forgery

1. Personal forgery

This can occur when someone writes a signature that is not their own, or alters a figure or some words. Common examples are forging signatures or altering the amounts on cheques, wills and insurance documents.

Cheques may be forged by blotting out words or numbers and replacing them with others, or by adding extra words or numbers to ones already there—for example, 'six' can be changed to 'sixty' and \$6 to \$60. This type of forgery can be detected if it can be shown that different inks have been used on different parts of the cheque or document.

2. Industrial forgery

Companies with famous brand names put a lot of money into research and development of their product. Their products are usually fairly expensive and of good quality. Industrial forgers copy products, often using poorer quality, cheaper materials, so that they can be sold very cheaply.

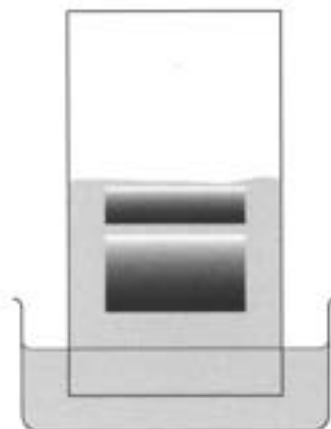
3. Forgery on metal

Many stolen goods—such as cars, bikes, keys, tools and guns—carry identification marks or serial numbers imprinted on them. Criminals try to remove these marks by filing them, in order to avoid discovery. Even though these marks may be no longer visible, the forensic scientist is still able to restore them. In activity 2.5 you will have the opportunity to do the same.

Using chromatography to identify a forgery

The inks used in pens are made up of a mixture of dyes or pigments. If these dyes can be separated, it can be seen from which colours the ink is made up. The ink from two different pens may seem the same colour when on paper, but be made up of different dyes. Chromatography is the method of separating the dyes.

It works like this. When a strip of absorbent paper is hung with one end in a beaker of water, the water will soak up into the paper. If an ink mark, that is soluble in water, is above the water surface, then as the water passes through the ink mark, some dyes in the ink mixture are attracted to the water and travel up the paper. The distance the pigment travels depends on the type of solvent used, and the weight and size of the dye molecules. The resulting pattern of colours is called a **chromatogram**.



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Questions

1. Give your own example of each of the three types of forgery.
2. Why is altering a cheque known as personal forgery?
3. What negative effects can industrial forgery have?
4. What information can be gained from a chromatogram?
5. Describe a possible crime scenario in which a forensic scientist might want to carry out a chromatography test.



The police need assistance in determining who might have written a ransom note left at a crime scene. The ransom note was written on white blotting paper with a black felt pen. A number of suspects have been identified and their pens seized. You need to figure out which one of the pens, if any, was used to write the ransom note.



Materials

- part of the ransom note
- numbered suspect black felt pens
- strips of filter or chromatography paper
- large container or trough of water
- pencil or stick to hold the strips of paper.



Procedure

1. Using one felt pen, draw a line on the paper strip 2 cm *above* the bottom of the paper. Label your paper strip in pencil at the top of the strip.
2. Attach your filter paper strip to the pencil or stick. Repeat with the other papers marked with the pens and also a sample of the ransom note.

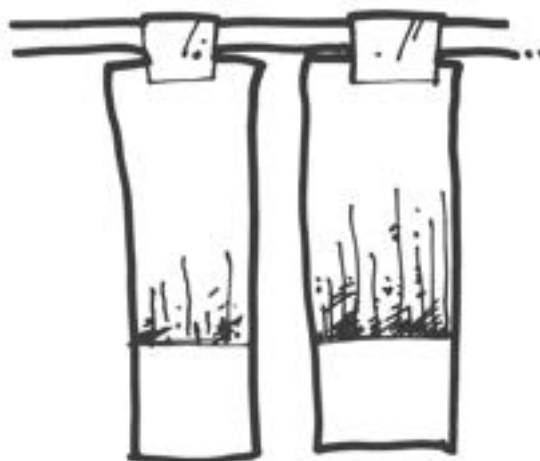
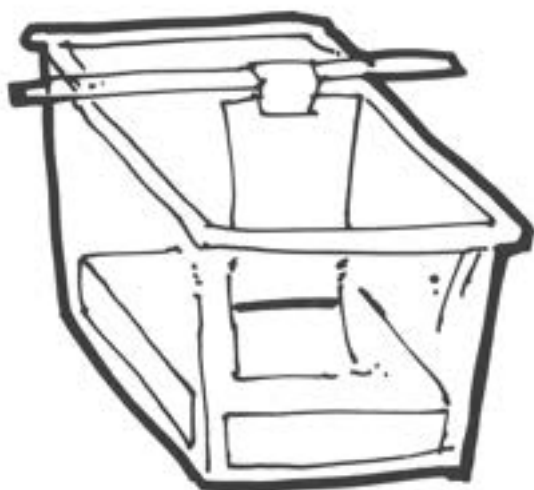


Activity: Who wrote the ransom note?

2 FORGERY



3. Lay the pencils with the paper strips across the container of water so that all the strips are hanging with the ink marks above the water surface.
4. Leave for ten minutes or until the water has travelled about halfway up the strip, carrying the dyes with it. Remove and compare the colour patterns.



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Questions

1. Which pen contained the greatest number of dyes?
2. Why did some dyes travel further than others?
3. Were there any inks that did not travel up the paper, or separate into other colours? If so, which ones, and why do you think this ink did not soak up with the water?
4. Which chromatograph was closest to the one from the ransom note?
5. What could you conclude about the pen that drew this line?
6. Why should the ink on the filter paper be above the water level?
7. Why were you asked to label your paper strip in pencil and at the top of the paper?
8. If the suspect's pen had not contained water-soluble ink (for example, a biro), water could not have been used. What other solvents may be used in chromatography?
9. How many English words that are not proper nouns can you make from the word 'CHROMATOGRAM'?



If words or numbers have been added to a cheque or other document it is likely that, although the inks will appear to be the same colour in daylight, they may show up as different colours under ultraviolet light. Only some inks will shine (fluoresce) in UV light. Only some inks will continue to shine after the UV light has been turned off (phosphoresce). A criminal attempting to increase the money value of a cheque, or change a will or other document, may think that they have succeeded by writing over the original in the same colour ink. The writing style might look expertly copied, but the forensic expert may be able to detect differences in the ink itself.



Materials

- UV lamp
- forged cheque (with extra zeros and letters added using a different brand pen to that used to write the original cheque).



Procedure

1. Look at the cheque you have been given in daylight. Try to see where a different coloured ink may have been used.
2. Now look at the same cheque under UV light. Try to identify some writing that may have been done in a different ink to the rest.



Questions

1. What changes were made on the cheque?
2. In what other ways could a cheque be forged?
3. How could you write the cheque you have inspected so as to make such a forgery difficult or impossible?
4. What other documents may a forger try to change? Give at least two examples.
5. Why is writing a cheque made out to 'Cash' a risky thing to do?
6. What is a blank cheque?
7. Under what circumstances do you think someone might write out a blank cheque?
8. Why is it foolish to give anyone, even a trusted family member, a blank cheque?
9. What would be a better way to overcome a situation where a blank cheque is sometimes used?





Metals are made up of tiny particles called atoms that are arranged in a regular and orderly pattern.

If a number is stamped on the metal, the layers of atoms below the mark are compressed (squashed together).

If the marks are filed away, the atoms left behind are still squashed where the mark was made; however, from above you cannot see the difference.

If a solution called Fry's reagent is used, it will change the appearance of the parts of the metal where the atoms are squashed. This will show up the mark that was filed away.

Fry's reagent is a corrosive acid that reacts with the atoms in the metal. In the area where the metal was punched, the atoms are closer together and so react with the acid faster than the surrounding area. Hence an etched pattern, showing a clear outline of the original marks, will appear. Fry's reagent is so corrosive that it must only be used by qualified science staff in a science lab. It should only be demonstrated.

The diagram on the right shows a registration mark stamped into a bicycle. Thieves may try to file the impression away and repaint the bike. Removal of the paint and treatment with a suitable etching reagent will show up the number that has been filed away.

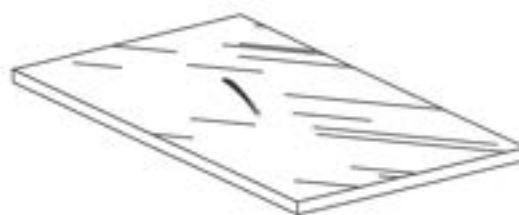
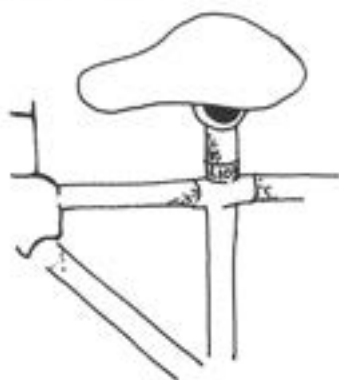
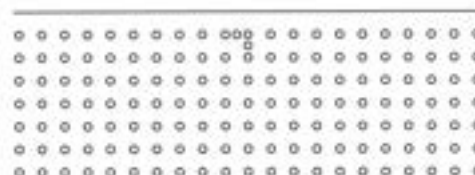
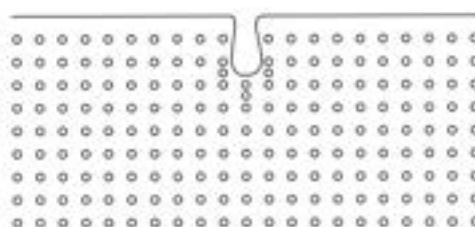
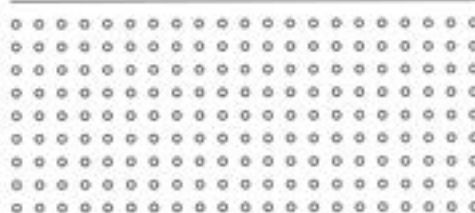


Fig. 13

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Questions

1. How can permanent numbers be attached to a bike or car engine?
2. Describe how a dishonest person removes such numbers.
3. Why is it so difficult to remove numbers completely from metals?
4. What effect does Fry's reagent have on a metal?
5. Why should the use of Fry's reagent only be demonstrated by qualified staff?



The solutions used in this experiment are very corrosive and should not be handled without gloves. It is best done as a teacher demonstration.



Materials

- a sample of metal (mild steel) that has had a punched mark removed by filing
- gloves
- safety glasses
- metal tongs
- cotton wool
- beaker of dilute ammonia large enough to fit the metal in
- Bunsen burner
- nitric acid (20%)
- acetone
- Fry's reagent (copper (II) chloride and hydrochloric acid solution).



Procedure

1. Wearing gloves and using tongs, dip some cotton wool in acetone and clean any grease from the metal (Fig. 14).
2. When the metal is clean, do not touch it again with your fingers (Fig. 15).



Fig. 15



Fig. 14

3. Using the tongs, gently heat the metal in a low flame for a few seconds. Do not let the metal glow red hot (Fig. 16).



Fig. 16

Activity: Detecting forgery on metal

2 FORGERY



4. Leave the metal to cool, then rub it for one minute with cotton wool dipped in Fry's reagent. Take care not to get any on you (Fig. 17).



Fig. 17

5. Dry the metal with clean cotton wool. Using tongs, dip a piece of clean cotton wool into nitric acid and rub the metal for one minute (Fig. 18).



Fig. 18

6. Dry the metal again with clean cotton wool (Fig. 19).

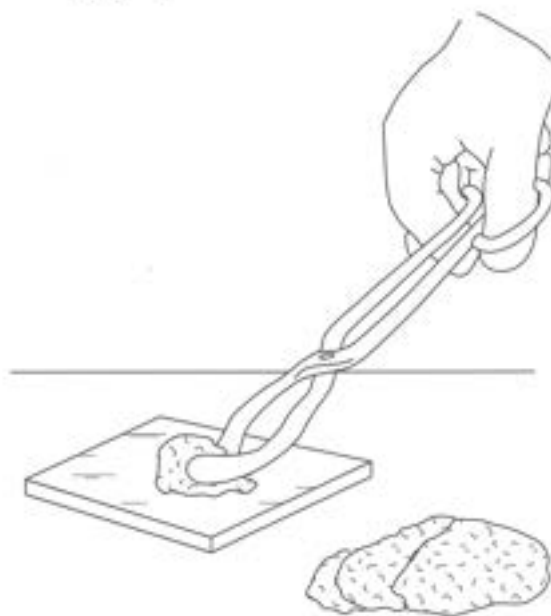


Fig. 19

Activity: Detecting forgery on metal

2 FORGERY



7. Repeat steps 4 to 6 until the forgery becomes clear (Fig. 20).

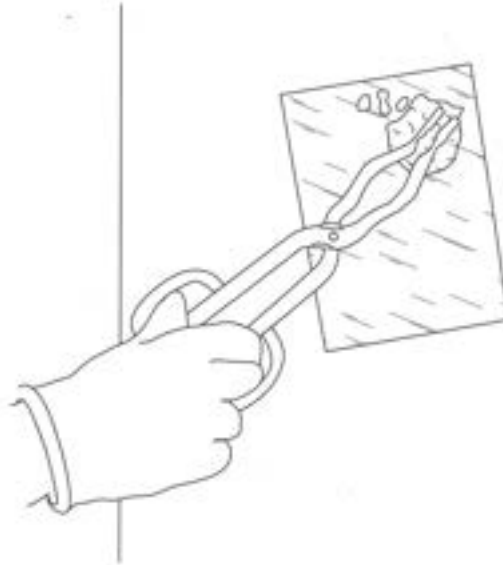


Fig. 20

8. Using tongs, wash the metal in ammonia in a beaker. Dry the metal with clean cotton wool (Fig.21).



Fig. 21

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Questions

1. Draw a picture of the marks found on the piece of metal.
2. Why is mild steel suggested as a suitable metal for this practical exercise?
3. Why are gloves as well as tongs used in this exercise?
4. What is the reason for not touching the metal with your hands in step 2?
5. Give some examples of forgeries where numbers might be filed off metal and replaced with new ones.



Handwriting experts are called in most often for these two common occasions:

1. when a signature is believed to be forged; and
2. when two samples of handwriting must be compared to check that the same person wrote them.

The fundamental principle common to all document examination involves a comparison of the suspect or disputed document with a known specimen of a person's signature or handwriting.

An individual's handwriting is personal and depends for its character on the interaction of the brain, eye and hand. It is affected by the person's physical and emotional state, the position in which they are writing and other circumstances—such as the type of pen or pencil—that may control the speed. A person may have several handwriting styles and several signatures. On the other hand, two identical signatures could look suspiciously like a photocopy or good quality tracing.

When comparing two samples of handwriting, a handwriting expert would particularly look at:

- the pattern of how high and how low the letters go
- the average space between the words or some letters
- the pressure and fluency of the writing
- the slope of the writing (it can be backhand '\', or normal '/' or upright '|')
- the relationship of the letters with the horizontal line
- the general style of writing (that is, cursive, printing or block letters)
- the style of writing certain letters—for example:

S or s, t or t, r or r, f or f, b or b, p or p

An expert in analysing handwriting can generally tell if a person is right-handed, and hence pulls the pen across the page, or left-handed, and pushes the pen across the page.

A right-hander pulls the pen so that it follows the wrist, while a left hander pushes the pen ahead of the wrist. These two very different actions put different emphasis on the down and up strokes in letter formation, even if the actual shape of each letter is very similar.



Fig. 22

Spaces comparison

A sheet of tracing paper can be placed over the signature and a dot drawn at the end of any letter that is not joined to the next one. This also includes spaces between words. The pattern shows where the pen of the writer was taken off the page. The gaps a person leaves are usually very individual.

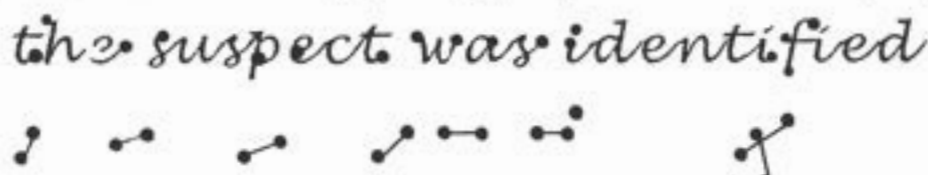


Fig. 23



Top-of-letter comparison

Another sheet of tracing paper is placed over a suspect signature, and a small dot is drawn on the highest point or points of each letter as shown below. When the tracing paper is removed, the dots are joined together to form a zigzag line. This can then be compared to the pattern formed from the genuine, or known signature.

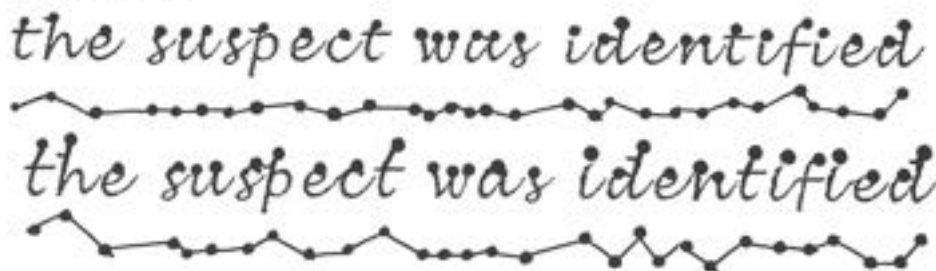


Fig. 24

While a casual observation of the letters themselves can detect no obvious differences, the two line graphs formed by the 'top-of-letter' technique show several clear differences.

Bottom-of-letter comparison

This is similar to the previous technique but, as the name suggests, a dot is placed on the lowest point or points of each letter. Again the two zigzag patterns of the known and suspect writing can be compared.

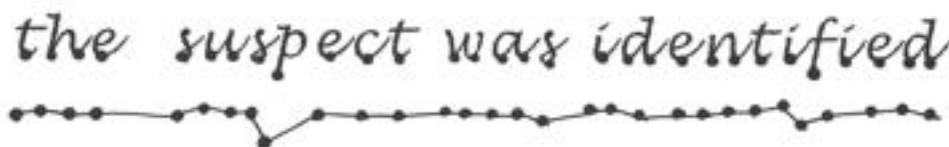


Fig. 25

Slope comparison

For this technique a line is drawn on the tracing paper, indicating each letter's slope. This too is a very individual characteristic.



Fig. 26



Questions

1. Describe two types of situations where a handwriting expert might be called in.
2. List seven things that a handwriting expert might look for when comparing two samples of handwriting.
3. What information about the writer can a handwriting expert gain from one sample?
4. Describe three different methods of comparison an expert might use in comparing handwriting.
5. How could these comparisons be done?



Before personal computers became so common, typewriters were used extensively. Formal and legal documents were either handwritten or typed. A criminal might have used a typewriter to write a threatening letter or ransom note in order to avoid their handwriting being recognised.

A forensic expert was often able to match a sample of typescript to a particular typewriter brand or model. They could vary in the way they typed '3' or '3', '4' or '4', where the centre point of 'M' or 'W' was and where the crossbars were on 'f' or 't'.

As a typewriter became older some letters would show wear in particular places. These errors would show up consistently every time the letter was typed throughout the whole document. Wear on these mechanical pre-electronic machines was another factor that created distinctions between them. Levers, springs, cogs and metal parts hitting and rubbing on each other all contributed to uneven wear and therefore a uniqueness in typing quality. This made many typewriters distinctive and, therefore, traceable. This meant that it could be quite easy to match a document with the exact typewriter on which it was written and hence eventually determine who wrote it.

The range in ink types was much larger than is now available for computers. There were greater numbers of manufacturers and supplier networks. Inks did not just come from a few global suppliers, but from many local sources, and each of these again had their own brand range and formulae. This made tracing easier because each ink was more distinctive.

In 1924 a 14-year-old boy was kidnapped and found murdered before any ransom money could be paid. An unusual pair of glasses was found at the site where the body was located and this led to one person whose alibi was very shaky—Nathan Leopold, a rich and brilliant 19-year-old student. Police also then questioned Leopold's close friend and fellow student, 18-year-old Richard Loeb. They both became prime suspects.

When a witness came forward, claiming to have seen two men dumping a typewriter in a local lagoon, an immediate search was made and the lagoon dragged. A typewriting machine was found and sent to the laboratory for forensic analysis. By then the police had matched the type on the ransom letters with the type on some of Richard Loeb's student essays. It was soon confirmed that the dumped typewriter was the machine on which both the ransom notes and Loeb's essays had been typed.

Both young men were found guilty of murder and kidnapping and were sentenced to life imprisonment.

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Questions

The piece of typewriter ribbon below was recovered from a suspect's rubbish bin. The partly destroyed message is a mirror image of what had been typed on to it.

if you don't leave \$10 000 then death will find

1. Can you work out what the message says exactly? (Hint: Do you have a mirror handy?)
2. What do you think the writer of this message might have been trying to accomplish?
3. Why would this piece of evidence not have existed if the suspect had used a computer?
4. Why are ransom notes produced on typewriters easier to trace than those done on computers?
5. Write down the dictionary meaning for the following words: extortion, ransom, kidnapping, alibi and suspect.



On Thursday, 28 September, businesswoman Ms Cynthia Shande was found on the floor of her upstairs hall. She had been poisoned and foul play was suspected. She was found clutching a handwritten will, leaving all her possessions to her cook, Mrs Muriel Thornton.

Ms Shande's nephew accused Mrs Thornton of forging the will and of murdering his maiden aunt for her possessions. The police were brought in.

You are a police forensic scientist specialising in disputed documents and handwriting, and you must compare the handwriting that has been found in Ms Shande's diary with that on the will (see Fig. 27).

*Wednesday 27th September
Several times today I did not
feel well. In the office it was
decided that rather than
persevere.*

extract from diary

*27 September, Wednesday
I hereby bequeath all my
worldly goods to my cook Mrs
Muriel Thornton. For several
years she has done a good job
of looking after me.*

extract from will

Fig. 27



Materials

- tracing paper
- pencil.



Procedure

1. Write down a list of letters that you can compare.
2. Draw up a table as shown below, and complete the comparisons listed.

Feature compared	From diary	From will
Slope of writing		
Shape of letter 's'		
Average gap between words		
Top-of-letter comparison		



Questions

1. Do you think that both samples of writing were written by Ms Shande? Explain in detail all the differences or similarities that have led you to make this conclusion.
2. What other information would be helpful in solving this crime?
3. In pairs, write two sentences, then swap with the other person and try to copy their writing. How easy is it to copy someone else's handwriting?



A blackmail letter, or a threatening message on note paper, may be the key to solving a crime. This may still be possible even if the actual document is not available to the police.

If the writing is on a special type of paper—such as recycled, wood-free or coated—it is possible to match it to paper from the same batch at a suspect's house or office. It may bear a unique watermark which can indicate where the paper was made or bought.

A note may have been torn from a page, or torn roughly into pieces so that the message could not be read again. Under a microscope the edges can be matched up, the pieces fitted together and the message revealed. An electron microscope can be used if even closer analysis is required.

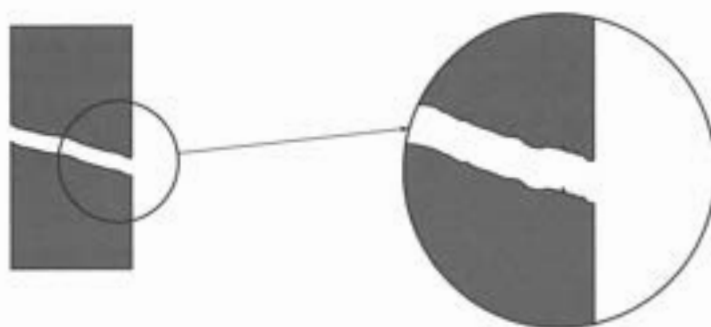


Fig. 28 Distinct tear patterns that may not be distinct under the naked eye may be able to be matched under magnification.

Many people press quite heavily when they write, especially when using a ballpoint pen. Such writing can leave indentations on any sheets underneath the paper being written on. The print can often be seen by allowing light to fall on it at a slant. By holding the paper at an angle near the light source, it is often quite easy to see these indentations. They can be photographed for a permanent record. Gentle rubbing with a soft pencil or some powder may also reveal the indentations by colouring the surface but leaving the indented parts uncoloured.

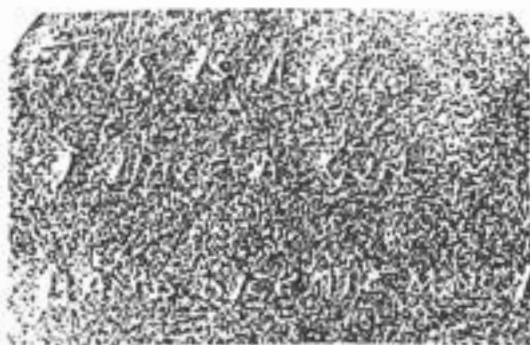


Fig. 29

Figure 29 shows how a criminal writing heavily on a notepad has left evidence of a message on the underlying sheet of paper for anyone to find.

The photograph was made by placing the flash gun at the side so that one side of the edges of the impressions cast a shadow while the opposite edge is lit up brightly. This allows the message to be read more easily, and is also a permanent record if the case should go to trial.



Electrostatic imaging to reveal indentations

Figures 30 and 31 are photographs of the same document. They are both impressions of a page that was below a sheet that had been written on by pen.

Figure 30 shows the page photographed under oblique light. Figure 31 is the same page, but it has been processed using an electrostatic imaging system. This provides the forensic scientist with more detail than the photograph.

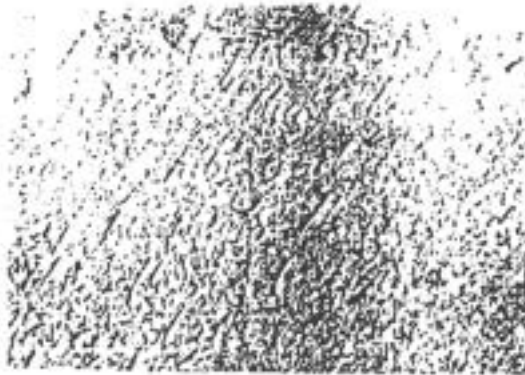


Fig. 30

w was in man or you
for I can't give you
less because I think he
vicious of him and he
is if he thinks I'm to
I have 118. 1 11

Fig. 31

How electrostatic imaging works

First, the suspect document is placed face upwards on the vacuum bed of the machine and covered with a thin transparent film. The film is pulled against the document by a vacuum pump. The indentions are then electrically charged before being covered with a fine powder of inked beads which attach to the indentions. They can be photographed or kept in place with adhesive film. The original document remains unmarked, which is important in case further tests need to be carried out.

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Questions

1. Explain how indented writing is produced.
2. What are two **simple** ways in which indented writing can be made visible?
3. How does electrostatic imaging work?
4. Give two reasons why electrostatic imaging is better than the ways mentioned in question 2.
5. Why is a photograph usually taken of the revealed message?
6. Write a short paragraph that describes how matching torn paper can be crucial evidence.



In this activity you are going to make your own indentations then try to read someone else's message by shading the top surface.



Materials

- sharp pencil or biro
- notepad with at least two sheets on it
- soft 'B' lead pencil or similar coloured pencil.



Procedure

1. With some layers of paper resting on a soft surface, write a message with a sharp pencil or a biro. You may like to try several messages, each time pressing with different amounts of force. Try resting the papers on different surfaces.



2. Remove the top sheet that you wrote on and dispose of this. Now inspect the sheet below. With the light coming from the side, can you read the writing?



Activity: Reading indented messages

2 FORGERY



3. Swap your message with a friend and see if you can read each other's notes. Very gently rub some pencil over the indentations. This should show up the message. Stick a sample in your science book.



4. Repeat this exercise, using a few different writing instruments on a piece of paper.
5. Repeat this exercise again, but this time use the same pen on different paper types.

?

Questions

1. How clearly could you read the messages?
2. What sort of things changed the clarity of the indentations?
3. Did you try resting the pages on both hard and soft surfaces?
Which type of surface causes the clearest indentations?
4. How did the type of pen that you used affect the impressions?
5. How did the type of paper that you used affect the impressions?
6. Write a paragraph describing a crime scene in which indented writing was used as the main evidence to solve it.



The Comparison projector

This works in a similar way to a comparison microscope. It is used to examine and compare two documents whose images are projected onto a screen. It allows the two images to be viewed side by side at the same time, just as with a comparison microscope.

However, the two images can also be superimposed, to check for very small differences. In this position, they can be flashed on the screen alternately, in which any differences will show up as flickering.

This instrument is particularly useful in picking up very small differences like misalignments and imperfections, because of its ability to enlarge the image many times on the screen.

Charred documents

With new techniques, even charred documents that have been burnt in a fire can be treated so that any writing or printing on them can be deciphered. Being very fragile and brittle, such papers have to be handled extremely carefully, otherwise the investigator could be left with a handful of useless black flakes.

If such documents are discovered, they are first photographed in the position in which they were found. Often these papers are twisted and curled up, and any attempt to flatten them would result in their destruction. To stabilise charred papers, they are first sprayed with a solution containing polyvinyl acetate, dissolved in acetone. This makes the paper more flexible, like a thin sheet of plastic. They can now be gently flattened and placed in a cotton-wool lined box, to be taken to the forensic laboratory.

In the laboratory, the sheets are carefully separated, then floated in one of a number of possible solutions. Under different lights, the writing or print will become visible so it can be photographed, and more easily read.

Another method that can be used, is to place the charred sheet between two unexposed photographic plates for two weeks. The result is claimed to be infallible, with a clear image being produced on the photographic plate every time. The disadvantage of this method is the time it takes to get the result.



Fig. 32



Papers

Most paper in use today is made from wood pulp. However, a few special papers still occasionally use cotton, linen or other plant fibres. Some security papers, like currency, use special additives to make forgeries easier to detect.

To make the expensive wood pulp go further, other fibres may be added, as well as a fine mineral chalk that makes the paper less porous and transparent.

Optical brighteners used as fluorocarbons can be added to make the paper appear whiter. This allows other colours to stand out more. The surface of the paper can also be treated in different ways such as heat rolling, or with coating of different chemicals.

All these various treatments allow different papers to be compared, and declared as possibly coming from the same source if the results are identical. These treatments have also been used to declare some documents, such as the Hitler diaries, as being fake, because the paper they were written on wasn't available in the 1940s. No one would attribute a work to Isaac Newton if it was written on synthetic paper in ball point ink.

Microscopic examination of paper fibres can often indicate the type of tree the pulp came from. If, as is common, more than one type is used, it can narrow the source of the paper down to only a few possible manufacturers.

Water marks

These have been used for centuries to identify the producer of the paper. Because they are regularly changed, a rough estimation of the time of manufacture is often possible.

The water mark is created by having slightly less fibres in these areas. This leaves a pale pattern visible when the paper is held up to the light. Because it is added during manufacture, it is impossible to copy at a later stage. Attempts have been made using oil and a print of the design, but these are easy to detect, as the edges of the copy are sharper than in the real water mark.

Another method used by criminals to copy a water mark, is to dampen some similar paper, then place it between two plates, where the water mark design has been raised on one plate. This can be detected under a microscope, however, as the pattern is compressed, whereas the original water mark has the same thickness as the rest of the paper.

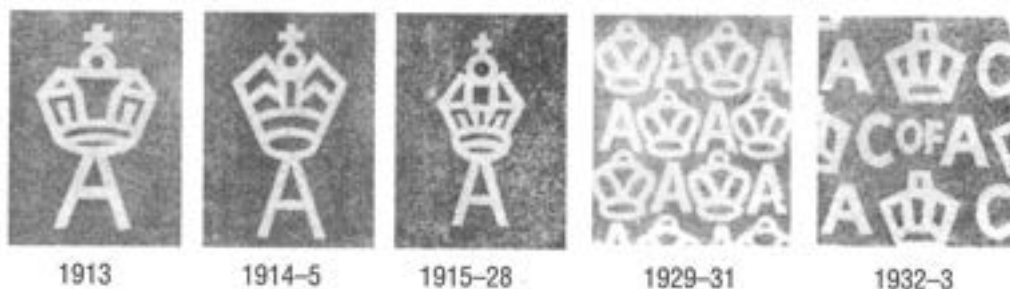


Fig. 33 Water marks used on paper from which Australia stamps were printed earlier in 20th Century.



Typewriter carbon ribbon

All modern typewriters and some computer printers use carbon ribbon to print what has been typed. The carbon ribbon moves from one plastic carrier past the keys to another carrier on the other side. The ribbon is coated in carbon so that as each key is struck, it transfers the shape of the letter onto the paper before moving on. If each section of ribbon is used only once, then it is possible by unwinding it and writing down all the letters that have been punched out in reverse order, to recreate the document, letter by letter.

Inks

Most inks consist of a dye that is dissolved in a solvent. Usually there are also other additives that help the ink stick to the paper, or help it to dry quickly. Most inks dry when the solvent has evaporated, and the dye has been absorbed into the paper.

Ancient inks were made from natural materials such as berries, bark, linseed oil and soot (carbon). For many centuries writing ink was made from an iron compound such as iron sulfate dissolved in tannin. Modern blue-black inks are still based on this mixture.

These are four main types of ink:

Ball-point ink which was first made in 1945. It contains dye in an oil base. Today it contains 45% dye, dissolved in polyethylene glycol.

Iron inks are described above. they are used mainly in fountain pens.

Carbon-black ink also called *Indian ink*, consists of carbon suspended in gum arabic (a gum from an acacia tree that grows in north Africa). It has also been in use for hundreds of years, and is still used mainly for drawing.

Dye-stuff ink consists of aniline (synthetic) dyes, gum-arabic, glycol, polyethylene and an acid. Like ball-point ink, it has only been used in the later half of the twentieth century.

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Questions

1. Describe two crimes for which a comparison projector might be used in the forensic laboratory.
2. In what way is the comparison projector similar to a comparison microscope? What are two differences in the way they are used?
3. Outline the steps taken to remove charred documents safely to the forensic laboratory.
4. What is the advantage of using the unexposed photographic plates method to make a copy of any print on a charred document? What is the main disadvantage of this method?
5. Explain how typewriter ribbon can be used to recreate a document that was printed on the instrument it was installed in.
6. What message was typed on the typewriter this ribbon was removed from?
..tirednulecrapehtecalP.eunevAtsriFdinateertShtimSforenrocehtnoxobrettehtotoG
7. What substance is paper mainly made of? What other substances may be used and why?
8. How authentic would you say a play written by Shakespeare was, if it was written on paper containing fluorocarbons, and the ink used contained polyethylene glycol? Explain your answer.
9. What is a water mark? What is it used for and how is it made?
10. Why is it so difficult for criminals to copy a water mark?



A number of years ago Britain's Admiralty Court was faced with a serious court battle. The supertanker Esso Chittagong and the freighter Thomas Everett had collided. The tanker owners accused the freighter of being at fault by sailing a collision course with their tanker. The freighter owners, however, insisted that their ship was going backwards and tendered the ship's log as proof.

In pencil was the number '3', indicating that the ship's whistle was blown three times. This is the maritime signal for sailing astern (i.e. backwards).

While the court was listening to the conflicting stories, a solicitor acting for the tanker owners, remembered a well known handwriting expert. He phoned Derek Davis and asked him to inspect the freighter's log.

Davis put the relevant page under an ultraviolet lamp. The paper around the number '3' was roughened, as if it had been rubbed. Hoping to find some remnants of the original pencil mark, Davis put the mark under infrared light. This only showed up a few lead fragments, not enough evidence to stand up in court.

Next Davis examined the page under the one that had the '3' on it. He placed a very thin sheet of plastic film over the page, then sprinkled some fine carbon powder on it. He added some 'glass beads' that are tiny particles like gold dust. When the page was shaken gently, the glass beads rolled the carbon powder into the indentations, making them visible under a microscope.

In the centre of the '3', Davis could see a tiny gap. The freighter's watchman had originally written a '2' (meaning turn to port), then later the lower half of the '2' had been rubbed out and replaced by a loop, turning the '2' into a '3'.

Davis showed this evidence to the judge, who found the tanker owners 'not guilty'. They were saved a bill of around \$2 million.

Derek Davis had given evidence in more than a thousand court cases in about fifty different countries. He has given his expert opinion on about 40 000 documents, including cheques, suicide notes, and certificates for professional qualifications.

A more unusual case was when he was called in to examine a cell door in an East London police station. On it was etched a murder threat using a nail file. Davis confirmed that it matched a particular prisoner's handwriting.

Some time after this firm sent him part of a toilet door panel. He was asked to identify which employee had scrawled vicious abuse about another employee's wife. He checked the writing against that of several suspects and identified the culprit.

'A writer can change slope, size, pressure and speed' said Davis, 'but lifelong writing habits are like fingerprints. They will nearly always find a person out, even in graffiti.'

A firm was suspicious about the size of a salesman's hotel claims. They sent Davis a bundle of signed receipts and he had no hesitation in deciding that only the first one was genuine. 'As for the rest, he'd invented hotels, had bill letterheads printed and filled in the details himself. The cashier's signatures, though disguised for variety, were all in the same handwriting – the salesman's.'

One of the tools regularly used by Davis is an exceptionally bright fibre optic light to scrutinise papers without burning them – 'mostly they are evidence and mustn't be damaged.' Another is a comparascope or comparison projector. This is used for checking the authenticity of printed documents by allowing magnified comparisons of two samples projected onto a screen.



An unusual case in India was brought to Davis when one man, wanting to re-marry, produced a birth certificate that his previous wife had been eaten by a Bengal tiger. Davis laid it under the comparascope side by side with a genuine Indian death certificate. Identical items appear as one under the machine's probing eye. In this case there was a flicker between the two images. 'The certificate was a superb fake.'

The will case that satisfied Davis the most, and which was the most unusual, was that of farmer Richard Stonier Powell, who died at 86 leaving his estate to the son of his brother Percy. Fourteen relatives thought that the will had been forged because the signature was shaky and not quite the same as his usual one. Although no-one ever writes an identical signature twice, some of the disparities were exceptional. Davis was able to show that Powell had had a number of strokes, some unrecorded, that had affected his signature. Davis convinced the court that Powell was getting over one such stroke when he signed his will. Percy's son duly inherited.

Derek Davis is unique in that he is self taught. All he ever wanted to be was a handwriting expert. From his own research, he published a technical paper showing that left handed writing can be distinguished from right handed. In 1969 this theory was put to the test in the courts. When he was called to the witness box he stunned the court by insisting that the accused was innocent because he was a right hander, and therefore 'pulled' his pen. The case made Davis's name famous far beyond the world of the police and the courts. Clients have queued for his services ever since.

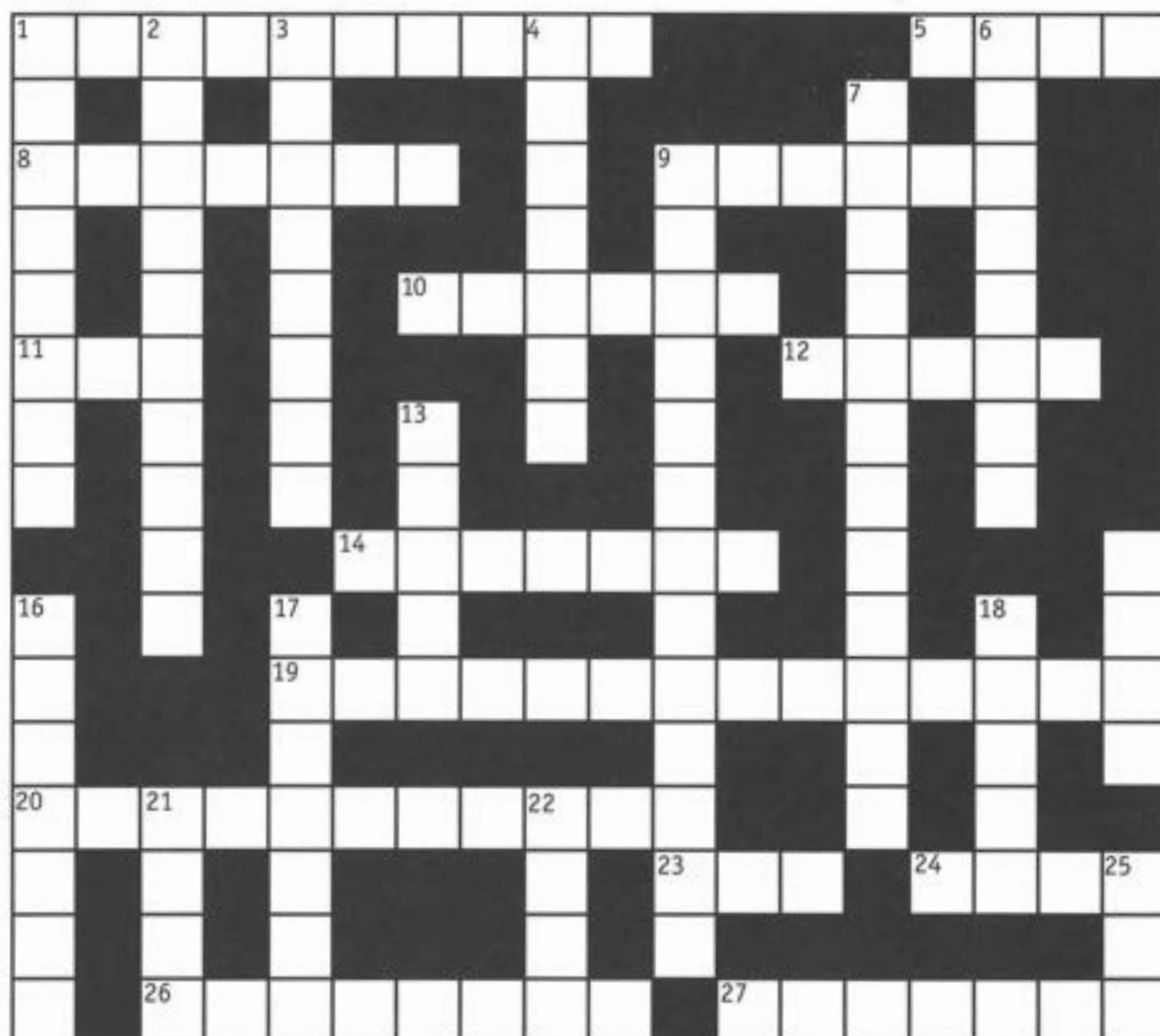
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Questions

1. What forgery was carried out on the freighter Thomas Everett?
2. What method was used to detect forgery carried out by someone on the freighter?
3. Who is Derek Davis?
4. Describe how Davis came to do the job he does now.
5. What case made Davis recognised as an expert in the field? Describe it briefly.
6. What was the most unusual sample of handwriting Davis had to compare?
7. Was he successful?
8. What sort of things can a handwriting expert use to identify a culprit?
9. Why would it be suspicious if 2 items being compared had identical signatures?
10. Why is Davis famous?



1. What type of forgeries would a handwriting expert be called in to investigate?
2. (a) If a specialist in disputed documents wanted to investigate a cheque that he or she suspected had been forged, which should be done first—inspect it under UV light, or treat the ink with liquid chromatography?
(b) Why?
3. Explain what a chromatogram is.
4. How is a chromatogram obtained?
5. (a) What causes the different dyes to travel up the filter paper when sat in a container of solvent during liquid chromatography?
(b) On what does the distance each dye travels depend?
6. Why don't all water-soluble black pens give the same chromatogram?
7. (a) What is the visible spectrum?
(b) Explain how natural white light can be broken down into its seven colours.
(c) Explain how infra-red radiation is different to ultraviolet radiation.
8. (a) Describe how a dishonest person can alter a cheque to defraud the cheque writer.
(b) In what way can the writer of a cheque prevent being cheated in this way?
(c) Explain how a scientist can use UV light to show up such a forgery.
9. Why should gloves be worn if using acetone, Fry's reagent or nitric acid?
10. In your science book, list at least five things that a handwriting expert will compare before deciding whether a particular signature or piece of writing is genuine.
11. What might be some reasons for making a handwriting expert suspicious if two signatures were identical?
12. What are three things an expert in comparing typewriting would look for in trying to determine if two typed documents were typed on the same typewriter?
13. Why have typewriting comparisons become less important in recent years?
14. Explain how indented writing is formed.
15. Briefly describe three ways in which a more permanent record can be made of a sample of indented writing.
16. Why are fingerprints such a useful tool in forensic science?
17. What does the term 'latent fingerprint' mean?
18. How have computers made fingerprinting a more useful tool?
19. How can a criminal avoid detection by changing their fingerprints?
20. Apart from sophisticated computer databases, what other improvements have been made in fingerprinting?





Across

1. A type of forgery in which famous brand name products are copied.
5. When indented writing is photographed, light should be from the
8. Something that is produced as an imitation with the intent to cheat or mislead.
9. Certain corrosive chemicals have done this to metals where atoms have been compressed together.
10. These should be worn when handling corrosive chemicals.
11. The colour next to orange in the spectrum.
12. A fluorescent substance a bluish glow when exposed to UV light.
14. A forger tried to pass a forged cheque at a busy railway
19. Liquid is used to spread out the coloured dyes used in some different pens.
20. This type of light has a shorter wavelength than visible light.
23. The substance found in pens that is a mixture of dyes or pigments.
24. Indented writing can be made visible by shading using a soft pencil.
26. The branch of forensic science concerned with forgeries is often called the documents branch.
27. The part of the electromagnetic spectrum that we can see.

Down

1. Type of radiation with a wavelength longer than visible light.
2. Famous British handwriting expert.
3. The name given to the seven colours of visible light.
4. A chemical used to remove grease from metal.
6. Type of writing found on a page below a page where a note was written using pressure.
7. The name given to the pattern of colours that spread out by a solvent carrying them up some filter paper.
9. A forensic scientist can make a very clear image of indented writing by using imaging.
13. A good solvent for the ink in water-soluble pens in liquid chromatography is.
15. The chemical used to reveal a removed serial number from metal is reagent.
16. To be suitable for liquid chromatography, the ink in the pen must be in the liquid.
17. To remove a serial number stamped on metal, a forger the whole top layer of metal away.
18. The between letters or words can help distinguish two samples of handwriting.
21. Filter paper strips must be so that the ink is above the solvent.
22. A sloping can be drawn on tracing paper to show the slant of a person's handwriting.
25. A different in pens of the same colour, can be revealed using chromatography.



1 Fingerprints

1-1 What are fingerprints?

1. Impression in soft material (if criminal touches fresh putty when climbing through a window); pattern left by stained fingers (if a criminal is bleeding, there may be blood on the fingers,) invisible fingerprints (may be left by a nervous criminal on anything that is handled).
2. Fingers, toes, palms, soles of feet, lips.
3. Sebaceous gland produces oil, sweat gland produces perspiration (sweat).
4. Invisible fingerprints are made by oils, sweat secretions (salts, urea, amino acids) some of which may be left behind when in contact with a hard surface.
6. Fingerprints form by the 5th month of pregnancy. They only change in size as the person grows, or if transfigured by a serious cut or burn
7. No one has the same fingerprint as anyone else.

1-2 Identifying fingerprints

2. The core is the centre of the pattern; the delta is the triangular shape where patterns meet.
3. Tented arch, radial and ulnar loop.
4. Whorl has a central core, and ridges turn completely about it. A central pocket loop has a core, but the rest of the pattern makes a sideways sweep to one side like a loop.
5. A fingerprint is a composite if it has more than 2 deltas.

1-3 Taking elimination fingerprints

1. Elimination prints would be clearer because they are taken carefully under police supervision.
2. If person is suspected of carrying out a crime, or to eliminate them if they have good reason to be at the crime scene.
3. Palm prints are taken because at some crime scenes only a palm print may be found.
4. (a) B;
(b) Suspect 4's thumbprint matched; No further evidence would be needed. This person may now be the 'prime' suspect.
(c) the third print may belong to another person who touched the cashbox, or be from the teacher's other thumb.

1-4 Activity: Taking your own fingerprints

4. The class distribution will most likely differ from that of the general population due to the very small numbers involved in one class.

1-6 How fingerprinting began

1. Timeline should include the following: ancient times—Chinese; 1684—Nehemiah Grew; 1684—Marcello Malpighi; 1858—Sir William James Herschel; 1880—Henry Faulds; 1890—

Francis Galton; 1892—Juan Vucetich; 1910—Edward Henry.

1-7 Matching fingerprints

1. A characteristic point is a small variation in the ridges (e.g. where they may cross, fork or leave a gap). Their presence and position are looked for when comparing two fingerprints.
3. The two fingerprints are an exact match because they have characteristics in common.
4. A crime scene fingerprint is usually round and only part of a whole figure. An elimination fingerprint is rolled, so appears square.

1-8 Latent fingerprints

1. A latent fingerprint is one that is not clearly visible. It has to be developed in the best way possible to make it visible.
2. Dusted fingerprints are lifted in case the photograph is not clear enough.
3. Iodine reacts with the oils in a fingerprint.
4. Iodine prints have to be sealed because iodine sublimates.
5. Superglue requires a humid environment in which to develop fingerprints.
8. The dust would stick to the sticky surface as well as to the fingerprint.
10. Blanket, woollen jumper, liquid surface, sand, cracked surface, pockmarked surface.

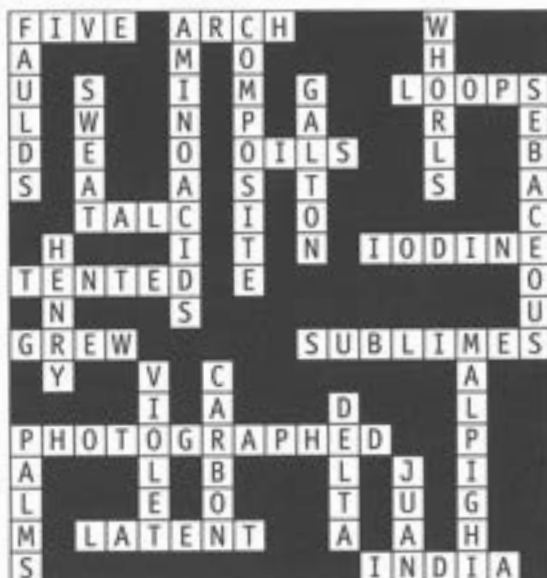
1-9 Activity: Dusting for fingerprints

1. Black, glossy, hard surface.
2. Dusting is not suitable because the dust would get into the fur or fabric.
3. Talcum powder is very fine and sticks to the smallest deposit of oil and sweat.
4. White or silver powder such as talc or aluminium powder would give the best contrast and so would photograph best.

1-10 Modern developments in fingerprinting

1. Using a computer saves a lot of time, and is more accurate. It results in a small number of fingerprints, in order of closeness to the fingerprint found at a crime scene, for the expert to inspect manually.
2. Lasers can detect old and fresh fingerprints and differentiate between them. They are completely non-destructive.
3. Since the 1970s several more methods of making a latent fingerprint visible have been developed. For example, fingerprints on skin and cloth can now be developed.
4. A fingerprint is unique to only one person. A computer database can link that person with a particular location.
5. More detailed computer analysis, better methods of developing fingerprints on rough and very absorbent surfaces.

1-11 Fingerprint crossword



2 Forgery

2-1 What is forgery?

1. Personal: parent's signature on note; industrial: bootleg music, using Olympic logos; metal: boat motors.
2. Because it involves the use of personal property such as a person's signature.
3. Inferior goods in the marketplace.
4. The different chemicals in an ink.

2-2 Activity: Who wrote the ransom note?

2. Different molecule sizes absorb into the paper at different rates.
5. It had a very similar chemical composition.
6. Ink would be lost directly to the water.
7. So that your writing would not interfere with the ink being tested.
8. Whatever breaks down the ink being tested.

2-3 Activity: Using UV light to show up a forgery

2. Changing the name of the person who is to receive the cheque, moving the position of decimal points, adding zeros, adding numbers before the decimal point, changing numbers.
3. Fill up all the space possible so no details can be added.
4. A will, a statutory declaration, deeds to property, insurance papers.
5. Anyone can bank this cheque.
6. Signed but no amounts added.
7. To someone they really trust who is to pay a tradesperson that day and the amount is not yet known.
8. It can be lost or stolen and used by the finder or thief.
9. Postal note, money order, credit card.

2-4 Forgery on metal

1. Fixed on to a plate, engraved, stamped.
2. Filing, removing the ID plate, restamping, painting.
3. The metal is affected below the surface.
4. It is corrosive and works differently on metal that has been worked on compared to untouched metal nearby.
5. It is corrosive.

2-5 Activity: Detecting forgery on metal

2. Easier to work with.
3. To protect the hands.
4. Sweat from your fingers can get on to the clean metal.
5. Video players, tools, hire equipment.

2-6 Handwriting comparisons

1. Contesting a will, insurance claims, personal cheques.
2. Slope comparison, bottom-of-letter comparison, top-of-letter comparison, spaces comparison, pressure, fluency, general style.
3. Right or left-handed; physical and emotional state.
4. As for question 2.
5. Superimposing samples to compare.

2-7 Case study: Typewriter ransom

1. '...if you don't leave \$10 000 then death will find...'
2. Blackmail.
3. No ink ribbons in a computer.
4. Difference in typing pressure, misaligned typewriter keys, generally less exact than a computer.

2-8 Activity: Detecting forgery by comparing handwriting

1. No. Slope, shape of letters, gaps between words and top of letter comparisons reveal differences.
2. The pen used, type of paper, access to paper, paper type.

2-9 Using paper or indentations on paper to detect fraud

1. Pressure from the pen through several sheets of paper.
2. Rubbing; shining a light at an angle.
3. Electrically charged ink attracted to indentations.
4. More detailed, less chance of damage to original.
5. It can be manipulated in terms of light, tone, etc. It is a permanent copy to take to court.

2-11

1. Any crime in which signatures or documents need to be compared e.g. suspected forgery of a will, or a cheque.
2. They are similar in that two things can be compared side by side. The comparison

projector allows superimposition and switching between two objects, while the comparison microscope does not.

3. The charred documents are sprayed with polyvinyl acetate, dissolved in a solution of acetone. They are then flattened and placed in a cotton-wool box, to allow safe transport to the laboratory.
4. A clearer image is produced. The disadvantage is that this takes a lot of time.
5. The letters on the ribbon are read backwards.
6. Go to the letter box on the corner of Smith Street and First Avenue. Place the parcel under it.
7. Wood pulp. Other substances may be cotton, linen, hemp, other plant fibres, and special chemical additives. These are used to create variety in look and texture of paper, or because the substance is cheaper than wood pulp.
8. The play would be a fake. These substances were not available in Shakespeare's time.
9. A pattern of paper where there is less fibre, usually used to identify manufacturer, or for security reasons (on currency). The water mark is made with a raised design on a surface on which the paper is resting during manufacture.
10. Difficult due to the water mark being made during manufacture.

2-12 Derek Davis - British handwriting expert

1. The ship's log recording the ships turn to port.
2. UV light showed something had been rubbed out.
3. A famous British handwriting expert.
4. His passionate interest in the subject led him to research and experiment until he became a recognised authority. He made his reputation by being scientifically accurate in court.
5. An Old Bailey case in 1969.
6. Richard Stonier Powell's.
7. Yes.
8. Slope, spacing, pressure, height, fluency.
9. Signatures, even from the same person, can change with mood, stress and age.
10. He is famous because of his success rate.

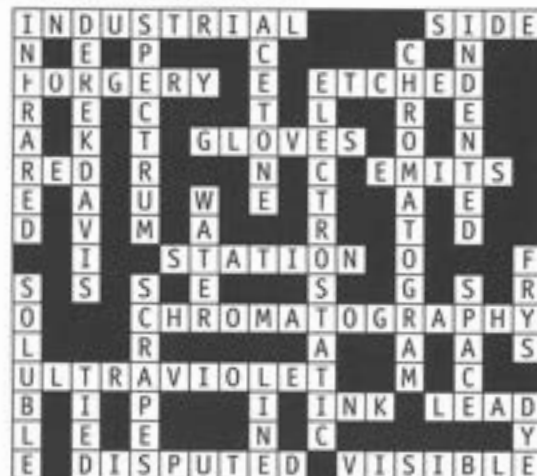
2-13 Forgery and fingerprint revision

1. Personal, industrial, on metal.
2. (a) Treat it under UV light.
(b) The original item is not damaged.
3. Chromatography is the method of separating dyes. A chromatogram is the paper with the colour spread on it.
4. Marking a test strip and suspending it in solvent.
5. (a) Dyes are attracted to the solvent and travel up the paper with it.
(b) The size and weight of the dye molecules.
6. Their inks are made up of different dyes.
7. (a) That part of the electromagnetic spectrum that is able to be detected by the human eye.
(b) Diffraction.
(c) Infra-red has a smaller wavelength.
8. (a) Alter the money amount.
(b) Leave no spaces that could be filled in.

(c) UV can detect things normal light may not.

9. They are very corrosive.
10. Space, slope, fluency, top-of-letter comparison, bottom-of-letter comparison.
11. The same person's handwriting does change, depending on stress, illness, age, mood.
12. Alignment of keys, shape of letters, pressure on to paper.
13. Computers have become more commonplace.
14. Pressure going through to the paper underneath the one being written on.
15. Photo, rubbing, electrostatic imaging.
16. They are low cost, quick to develop and usually easy to record and present to the court.
17. These are prints that cannot be seen easily by the naked eye but show up using various forensic techniques.
18. Computer databases of prints in countries around the world can be almost instantaneously accessed by police forces and crime prevention agencies. They also allow a wide range of image enhancement and comparison possibilities.
19. They can't because the resultant scar tissue can be just as distinctive as the original print.
20. The laboratory equipment and the chemicals used are more sophisticated. Methods for developing prints have become better suited to the task.

2-14 Forgery Crossword



16. They are low cost, quick to develop and usually easy to record and present to the court.
17. These are prints that cannot be seen easily by the naked eye but show up using various forensic techniques.
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ISBN: 978-1-92571-070-0



9 781925 710700

Emerald City Books
An imprint of Blake Education Pty Ltd.
A.B.N. 074 266 023
Locked Bag 2022
Glebe NSW 2037

